

SKILL NEXIS

Week 4 Capstone Project

Global Superstore Sales Data Analysis

Project Requirement	Status
Exploratory Data Analysis	Completed
Regression Model	Completed
Power BI Dashboard	Completed
Findings & Recommendations	Completed

This report presents the analysis, business findings, recommendations, regression model summary, and Power BI dashboard overview for the Week 4 capstone project.

1. Project Overview

The project analyzes Global Superstore sales data using Python/Pandas and Power BI. The work includes exploratory data analysis, business performance analysis, a regression model to predict sales, and an interactive dashboard for communicating key insights.

2. Key Performance Indicators

Metric	Value
Total Sales	118.73M
Total Profit	16.89M
Total Units Sold	1.13M
Sales–Profit Correlation	Approximately 0.805
Regression R ²	Approximately 0.651

3. Exploratory Data Analysis

- The dataset was loaded from Excel using Pandas and inspected using initial rows, data types, null-value checks, and descriptive statistics.
- Missing values in the Discount Band field were handled using the label “Unknown”.
- The analysis examined performance by product, country, segment, year, month, and profit.

4. Key Findings

Overall performance: Total sales were approximately 118.73M, total profit was 16.89M, and total units sold were approximately 1.13M.

Product: Paseo was the top-selling product at approximately 33.01M in sales and approximately 4.80M in profit. Carretera had the lowest sales among the six products.

Country: The United States of America generated the highest sales at approximately 25.03M. Mexico generated the lowest sales among the five countries at approximately 20.95M.

Segment: Government generated the highest sales at approximately 52.50M, followed by Small Business at approximately 42.43M. Channel Partners generated the lowest sales.

Yearly trend: Sales increased substantially from approximately 26.42M in 2013 to approximately 92.31M in 2014.

Monthly trend: October had the highest monthly sales at approximately 21.67M, followed by December at approximately 17.37M. March had the lowest at approximately 5.59M.

Correlation: Sales and profit showed a strong positive correlation of approximately 0.805.

5. Regression Model

A Linear Regression model was created to predict Sales using Profit as the input feature. The model used a train/test split with random_state=42. The test-set R^2 was approximately 0.651.

The fitted model was approximately: Predicted Sales = 61,246 + 4.551 × Profit.

The R^2 value indicates that the model explains about 65% of the variation in the test-set sales values. The model is presented as an analytical exercise for the capstone.

6. Power BI Dashboard

The completed Power BI dashboard provides a visual summary of the main business metrics and trends. It contains Total Sales, Total Profit, Total Units Sold, Sales by Product, Sales by Country, Sales by Segment, Monthly Sales Trend, and a Country slicer.

The Month Name field was sorted using Month Number so that the monthly trend is displayed in chronological order.

7. Recommendations

- Focus on high-performing products such as Paseo while investigating lower-performing products such as Carretera.
- Maintain strong sales activity in the United States and investigate opportunities to improve performance in Mexico.
- Prioritize the Government and Small Business segments because they generate the largest sales volumes.
- Review the drivers behind the large increase in sales from 2013 to 2014 and identify strategies that can be reused.
- Prepare inventory and marketing plans for high-sales periods, particularly October and December.
- Track profit together with sales when evaluating growth so that revenue growth remains commercially sustainable.

8. Final Deliverables

- Skill_Nexis_Week_4_Capstone.pbix — Power BI dashboard
- Week-4-Capstone-Project.ipynb — Python EDA and regression notebook
- Week-4-Sales-Data.xlsx — dataset
- Skill_Nexis_Week_4_Final_Report_Clean.pdf — project explanation report

Prepared for the Skill Nexis Week 4 Capstone Project.